Connected audio path (1)

File: sheets/10-audio-path-1.kicad_sch

Connected audio path (2)

File: sheets/10-audio-path-2.kicad_sch

Connected audio path (3)

File: sheets/10-audio-path-3.kicad_sch

Connected audio path (4)

File: sheets/10-audio-path-4.kicad_sch

Envelope core and converters (1)

File: sheets/20-envelope-core-1.kicad_sch

Envelope core and converters (2)

File: sheets/20-envelope-core-2.kicad_sch

Envelope core and converters (3)

le: sheets/20-envelope-core-3.kicad_sch

Envelope core and converters (4)

File: sheets/20-envelope-core-4.kicad_sch

Envelope core and converters (5)

File: sheets/20-envelope-core-5.kicad_sch

Envelope core and converters (6)

File: sheets/20-envelope-core-6.kicad_sch

Envelope core and converters (7)

File: sheets/20-envelope-core-7.kicad_sch

Envelope core and converters (8)

File: sheets/20-envelope-core-8.kicad_sch

Envelope core and converters (9)

le: sheets/20-envelope-core-9.kicad_sch

Envelope core and converters (10)

File: sheets/20-envelope-core-10.kicad_sch

Gate, latch, comparator, segment logic (1)

File: sheets/30-gate-latch-1.kicad_sch

Gate, latch, comparator, segment logic (2)

File: sheets/30-gate-latch-2.kicad_sch

Gate, latch, comparator, segment logic (3)

File: sheets/30-gate-latch-3.kicad_sch

Gate, latch, comparator, segment logic (4)

File: sheets/30-gate-latch-4.kicad_sch

ate, latch, comparator, segment logic (5)

File: sheets/30-gate-latch-5.kicad_sch

Control summer and current source (1)

File: sheets/40-control-1.kicad_sch

Control summer and current source (2)

File: sheets/40-control-2.kicad_sch

Control summer and current source (3)

File: sheets/40-control-3.kicad_sch

Control summer and current source (4)

File: sheets/40-control-4.kicad_sch

Control summer and current source (5)

File: sheets/40-control-5.kicad_sch

Power, reference, string, followers (1)

File: sheets/50-references-1.kicad_sch

Power, reference, string, followers (2)

File: sheets/50-references-2.kicad_sch

Power, reference, string, followers (3)

File: sheets/50-references-3.kicad_sch

Power, reference, string, followers (4)

File: sheets/50-references-4.kicad_sch

Selection position

File: sheets/60-selection.kicad_sch

Controls, ENV_OUT, I/O, test points (1)

File: sheets/70-controls-io-1.kicad_sch

Controls, ENV_OUT, I/O, test points (2)

File: sheets/70-controls-io-2.kicad_sch

Controls, ENV_OUT, I/O, test points (3)

File: sheets/70-controls-io-3.kicad_sch

Controls, ENV_OUT, I/O, test points (4)

File: sheets/70-controls-io-4.kicad_sch

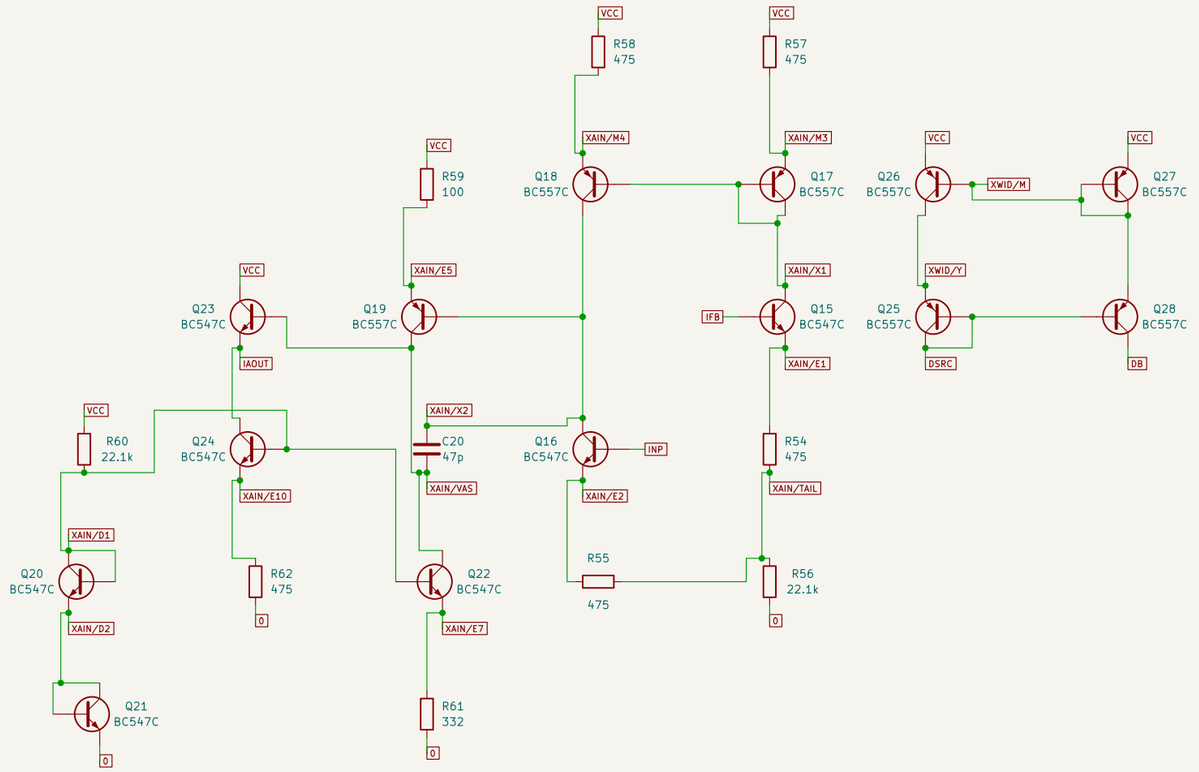
Mamut VCA-1 A0 discrete THT (overview)
AUDIO_IN -> input amplifier (x6) -> linearized cascode cell -> transimpedance buffer -> AUDIO_OUT
GATE_IN -> Schmitt -> edge set / latch / peak comparator -> segment logic -> target switches and steered converter currents
One follower drives C_T toward VOVER, SUSTAIN, or VREL; JFET buffer gives ENV
ENV, LEVEL_CV, OFFSET -> translinear product and summer -> clamp -> V-to-I -> I_CTRL (cell tail)
Reference: 1N829A, gain 1.25 (REF trim), one string: VOVER VREF5 VCASC VBIAS VCB VB0 VENV0 VREL VHI
Trims: FEEDTHROUGH, GAIN, TIME_SCALE, TIME_SPAN, FOLLOWER_OFFSET, REF. Physical tests NOT RUN.
Each page draws its local nets as wires; a global label names each net and connects it to the other pages.

Sheet: /
File: mamut-vca-1-a0-tht.kicad_sch

Title: Mamut VCA-1 A0 / overview

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 1/34

Connected audio path (1) — units XAIN, XWID



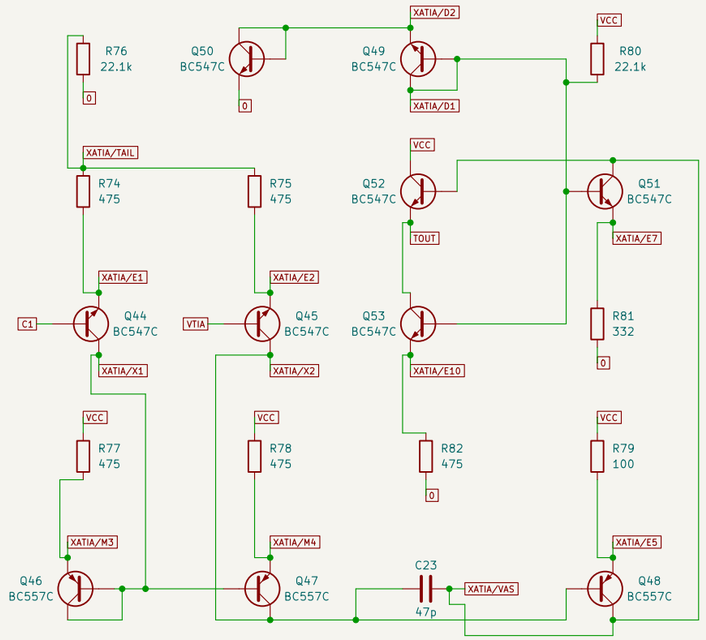
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Connected audio path (1)/
File: 10-audio-path-1.kicad_sch

Title: Mamut VCA-1 A0 / Connected audio path (1)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 2/34

Connected audio path (2) — units XATIA



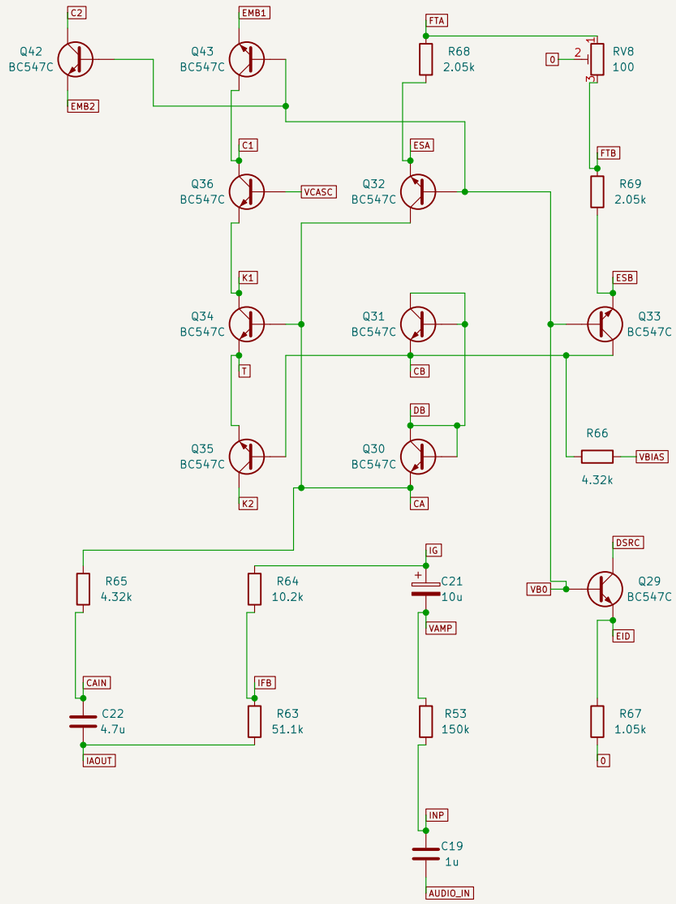
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Connected audio path (2)/
File: 10-audio-path-2.kicad_sch

Title: Mamut VCA-1 A0 / Connected audio path (2)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 3/34

Connected audio path (3)



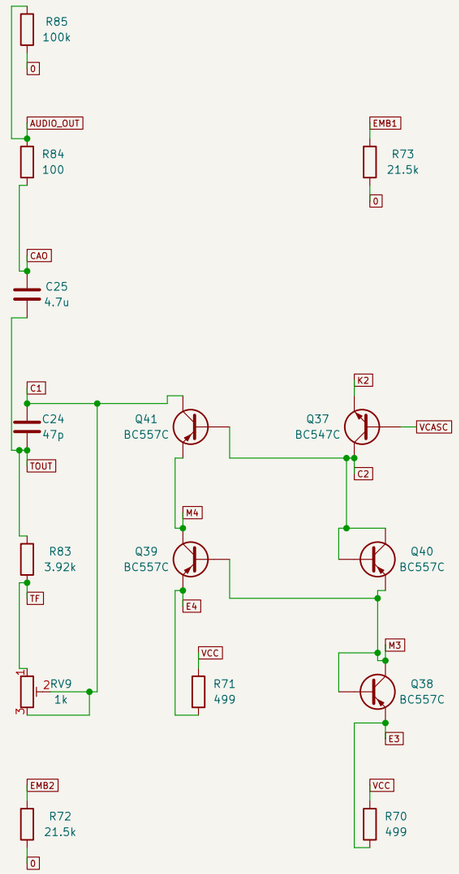
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Connected audio path (3)/
File: 10-audio-path-3.kicad_sch

Title: Mamut VCA-1 A0 / Connected audio path (3)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 4/34

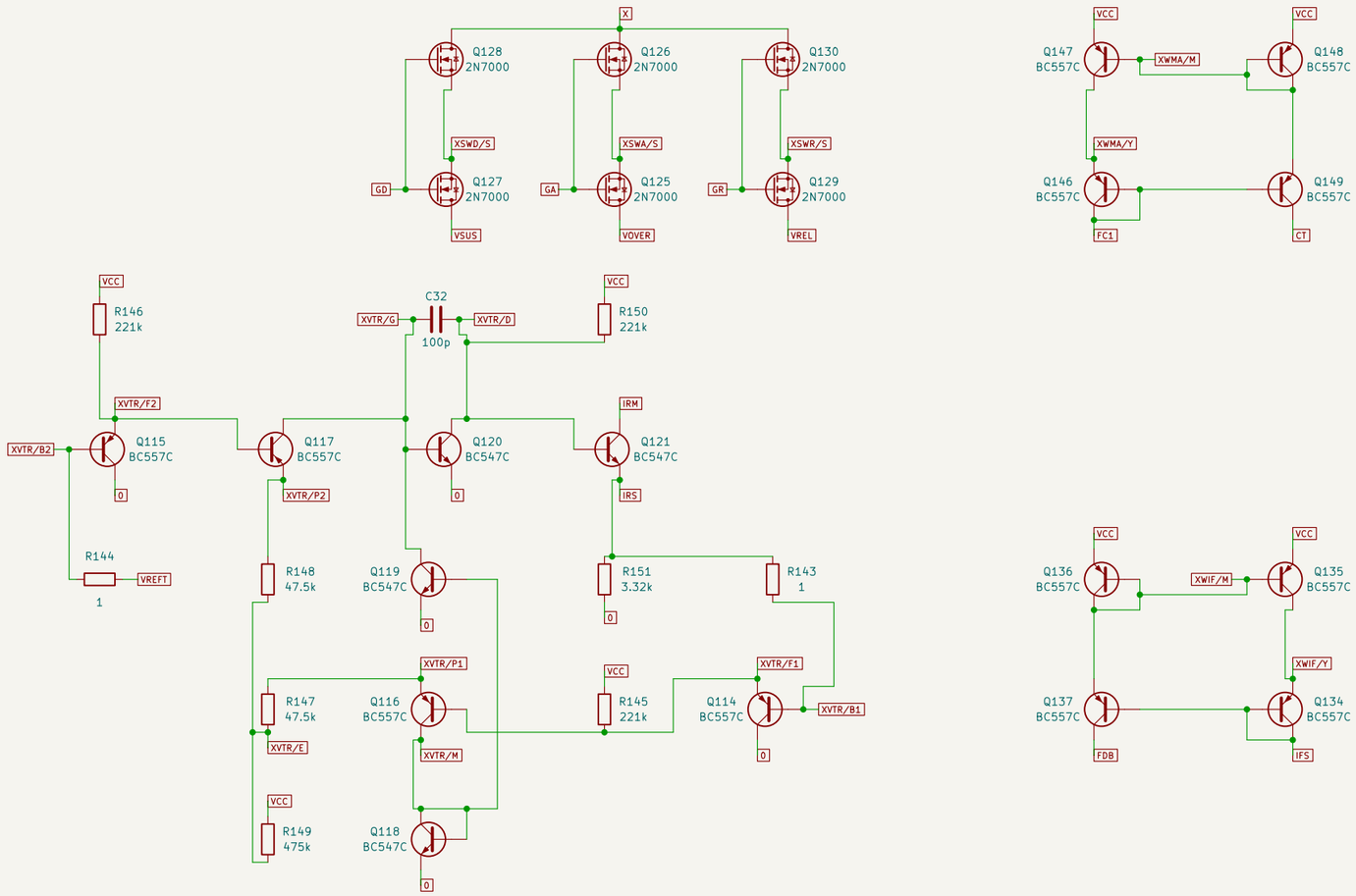
Connected audio path (4)



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Connected audio path (4)/		
File: 10-audio-path-4.kicad_sch		
Title: Mamut VCA-1 A0 / Connected audio path (4)		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 5/34

Envelope core and converters (1) — units XVTR, XSWA, XSWD, XSWR, XWIF, XWMA



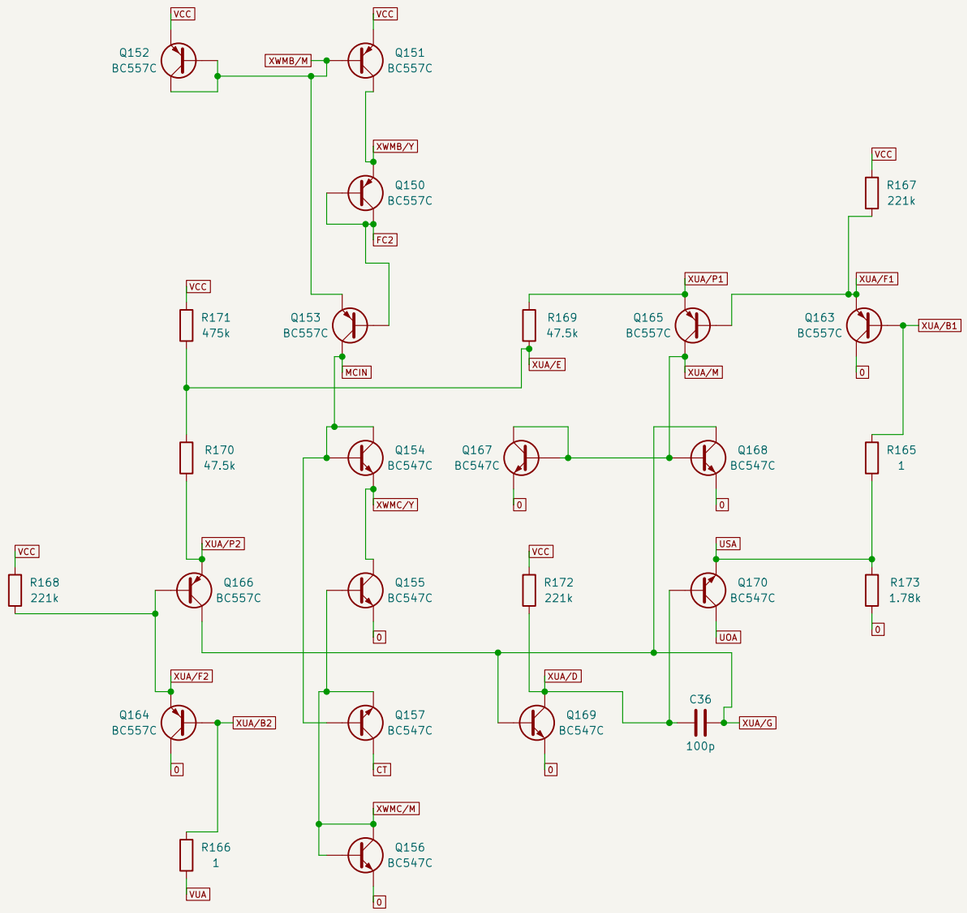
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (1)/
File: 20-envelope-core-1.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (1)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 6/34

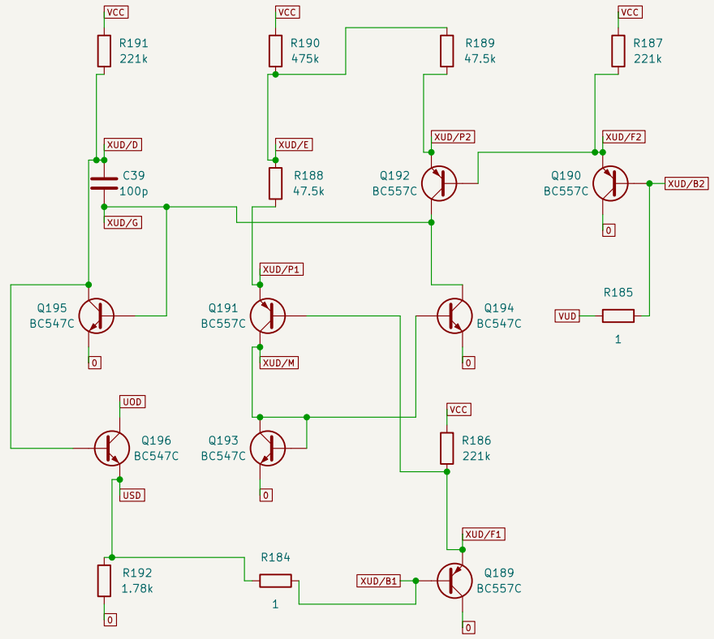
Envelope core and converters (2) — units XWMB, XWMC, XUA



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENVO, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (2)/	
File: 20-envelope-core-2.kicad_sch	
Title: Mamut VCA-1 A0 / Envelope core and converters (2)	
Size: A3	Date:
KiCad E.D.A. 10.0.7	**Rev: A0**
	Id: 7/34

Envelope core and converters (3) — units XUD



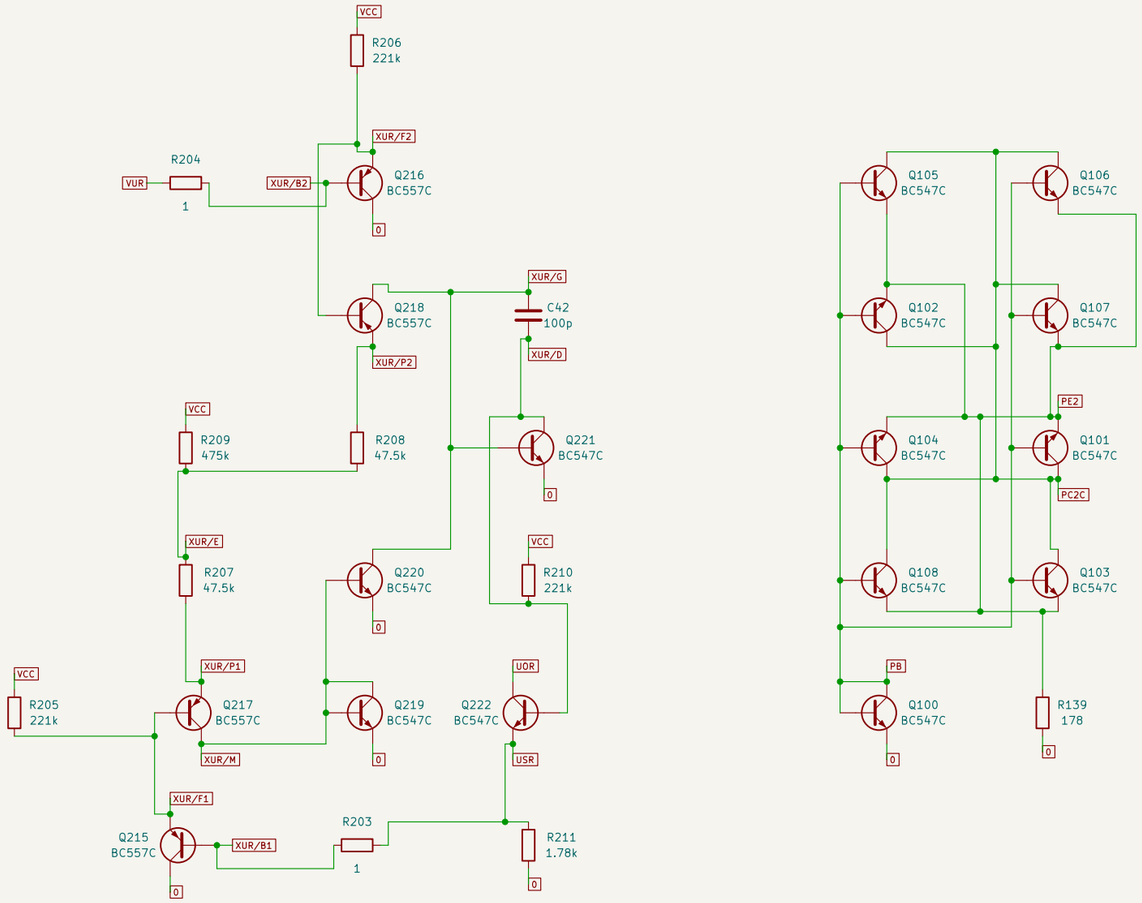
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (3)/
File: 20-envelope-core-3.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (3)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 8/34

Envelope core and converters (4) — units XUR



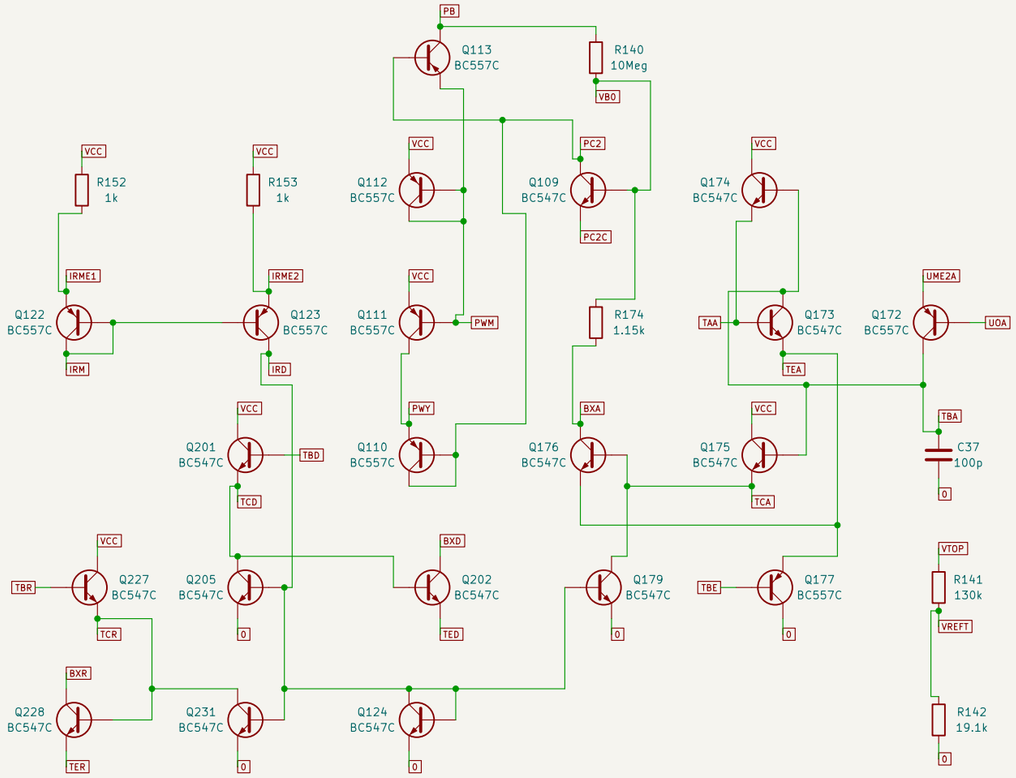
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (4)/
File: 20-envelope-core-4.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (4)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 9/34

Envelope core and converters (5)



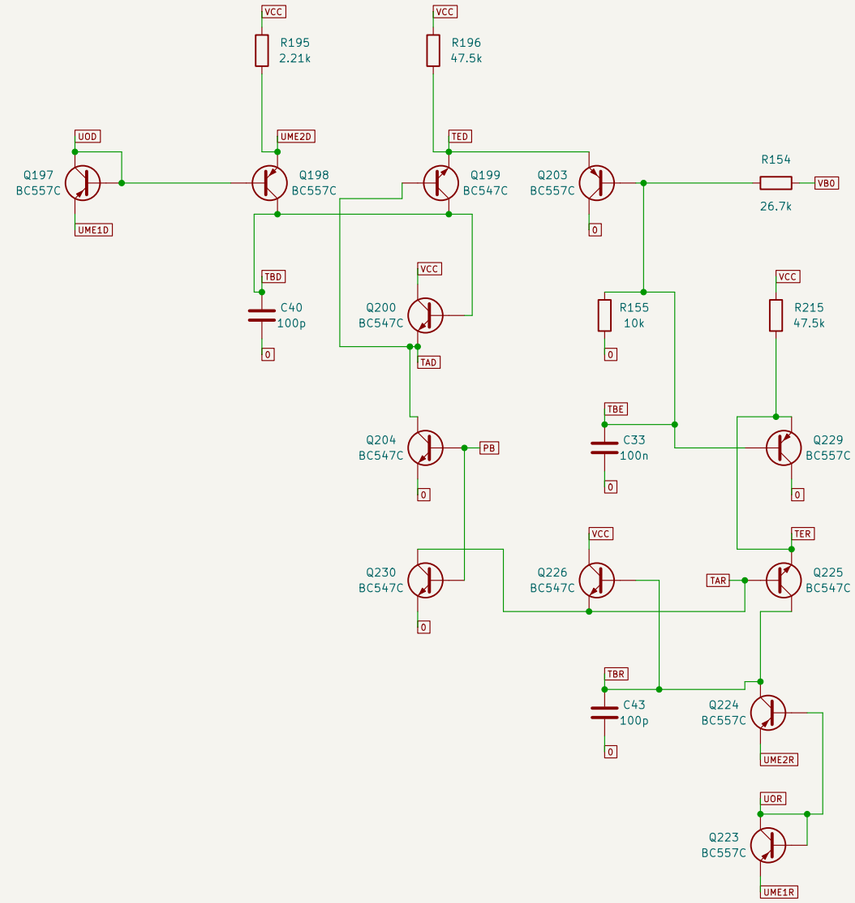
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (5)/
File: 20-envelope-core-5.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (5)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 10/34

Envelope core and converters (6)



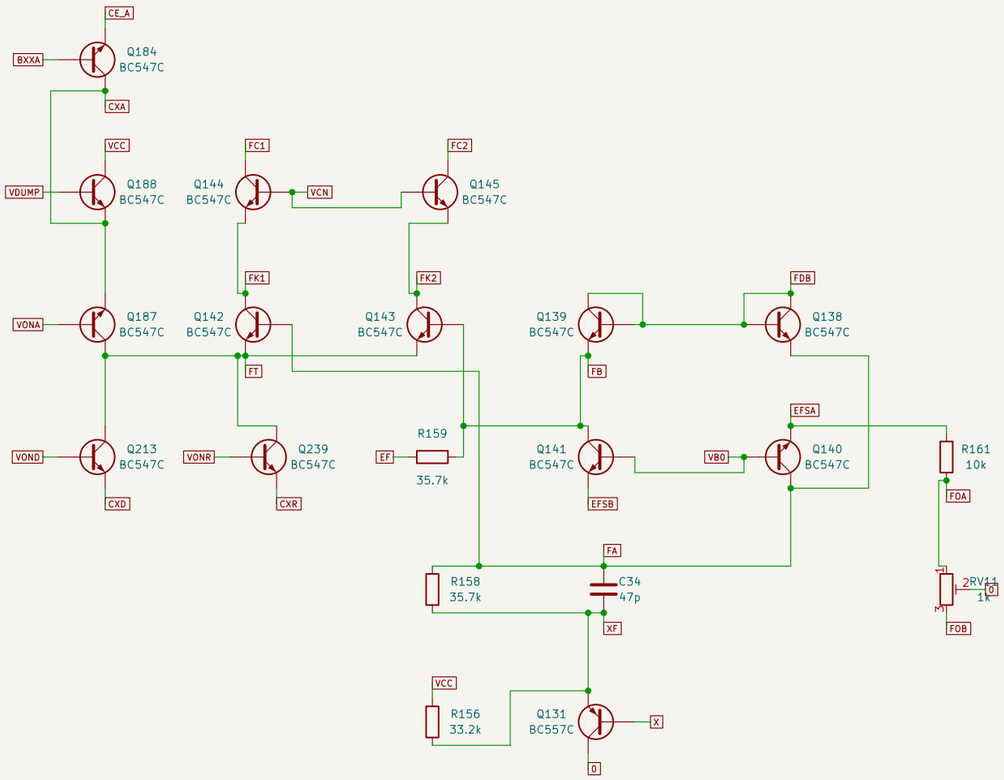
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (6)/
File: 20-envelope-core-6.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (6)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 11/34

Envelope core and converters (7)



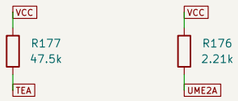
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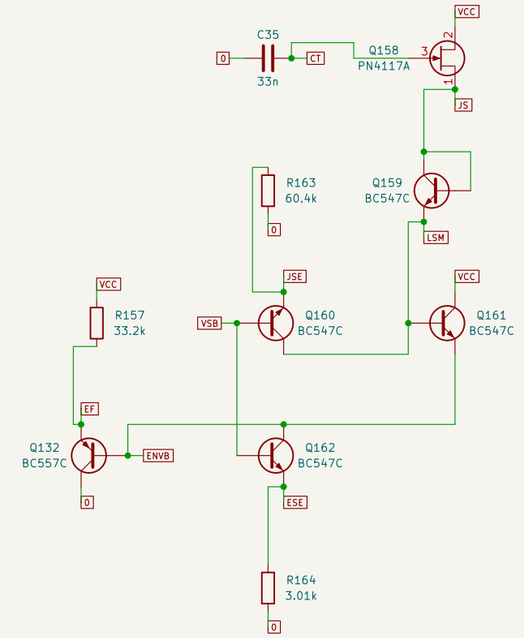
Sheet: /Envelope core and converters (7)/
File: 20-envelope-core-7.kicad_sch

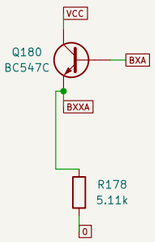
Title: Mamut VCA-1 A0 / Envelope core and converters (7)

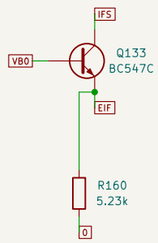
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 12/34

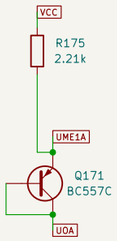
Envelope core and converters (8)

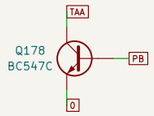














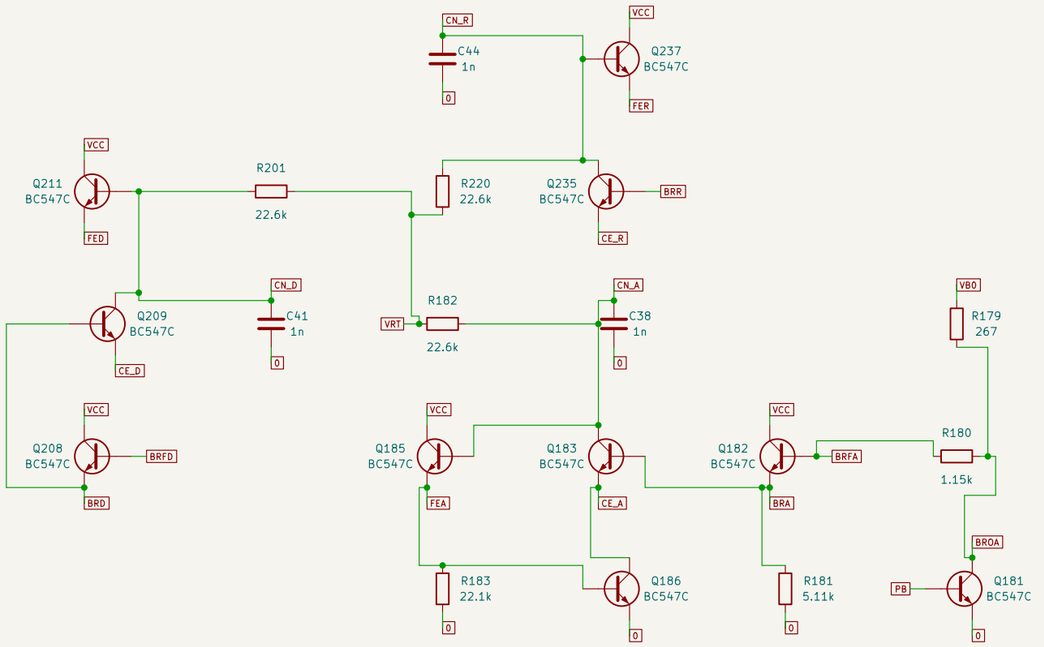
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (8)/
File: 20-envelope-core-8.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (8)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 13/34

Envelope core and converters (9)



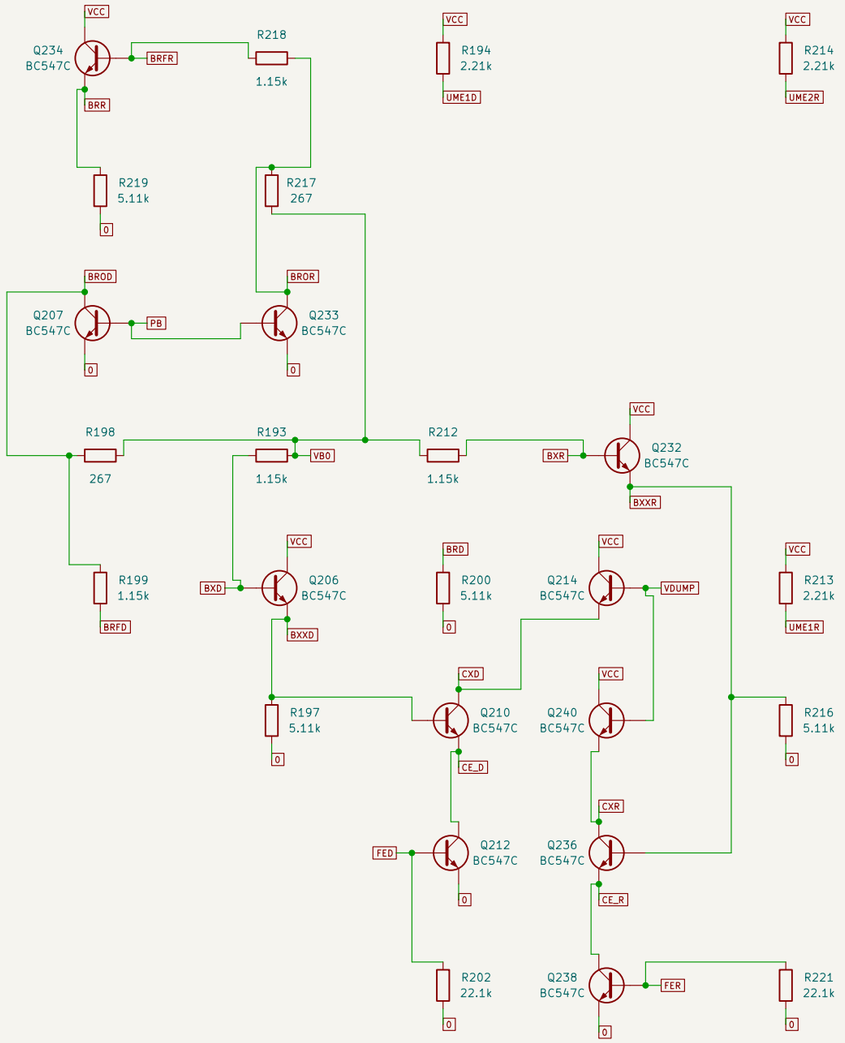
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (9)/
File: 20-envelope-core-9.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (9)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 14/34

Envelope core and converters (10)



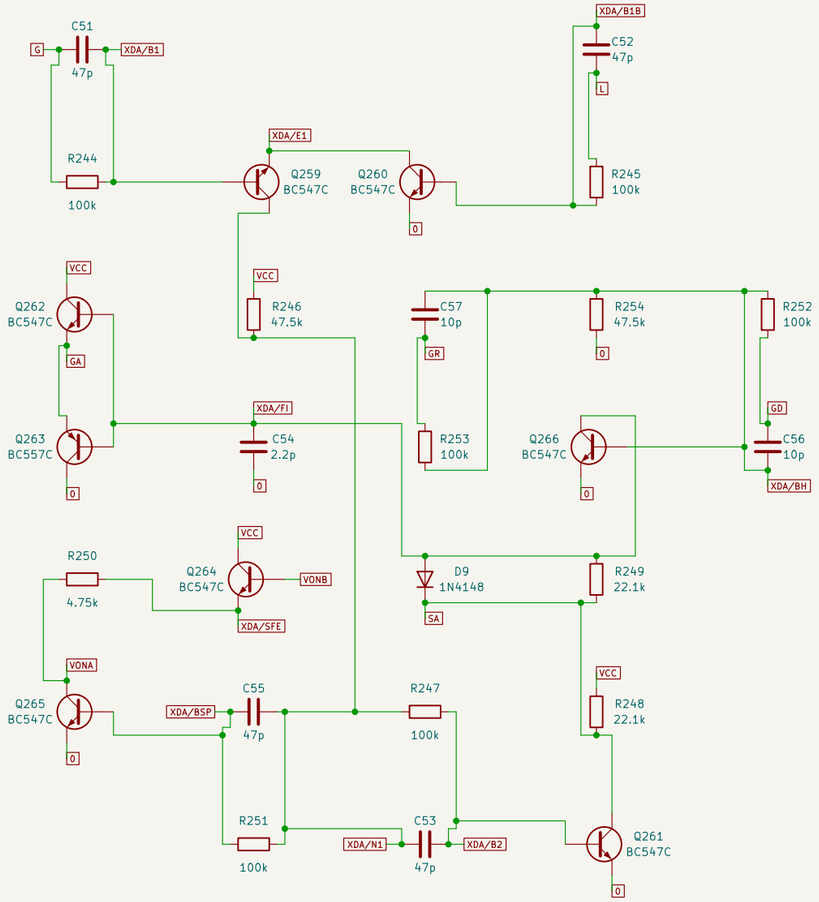
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Envelope core and converters (10)/
File: 20-envelope-core-10.kicad_sch

Title: Mamut VCA-1 A0 / Envelope core and converters (10)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 15/34

Gate, latch, comparator, segment logic (1) — units XDA



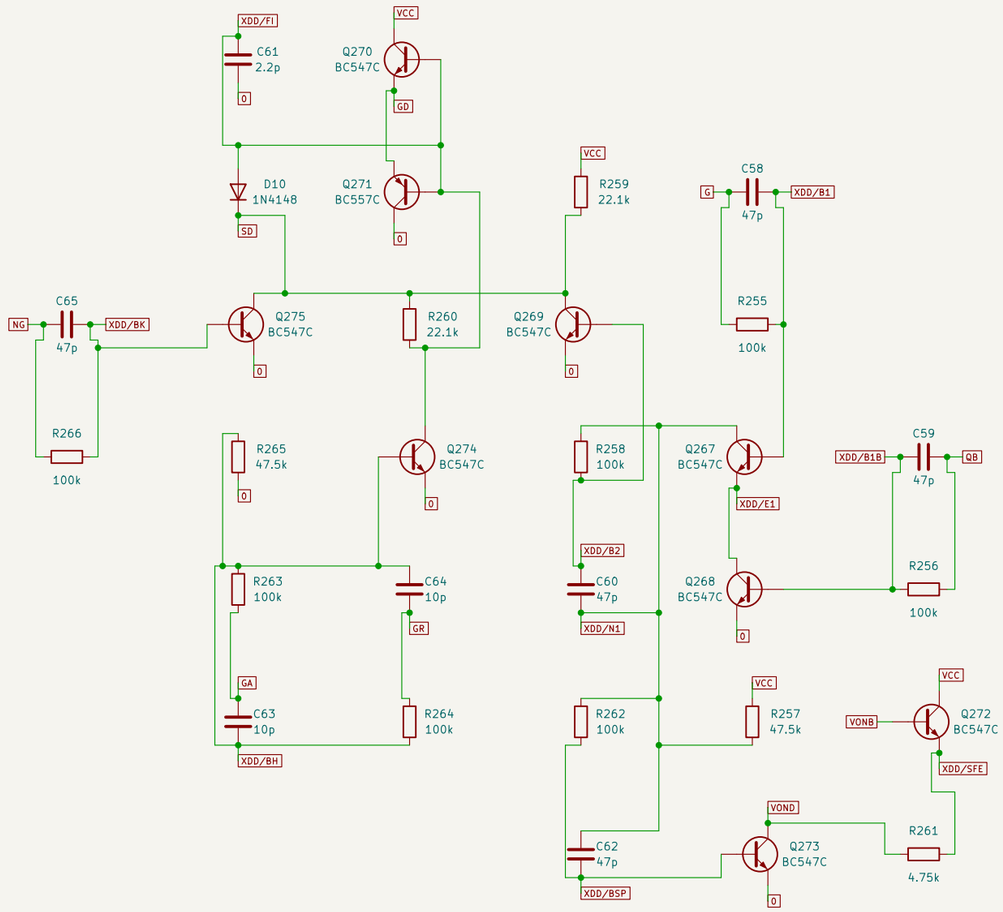
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Gate, latch, comparator, segment logic (1)/
File: 30-gate-latch-1.kicad_sch

Title: Mamut VCA-1 A0 / Gate, latch, comparator, segment logic (1)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 16/34

Gate, latch, comparator, segment logic (2) – units XDD



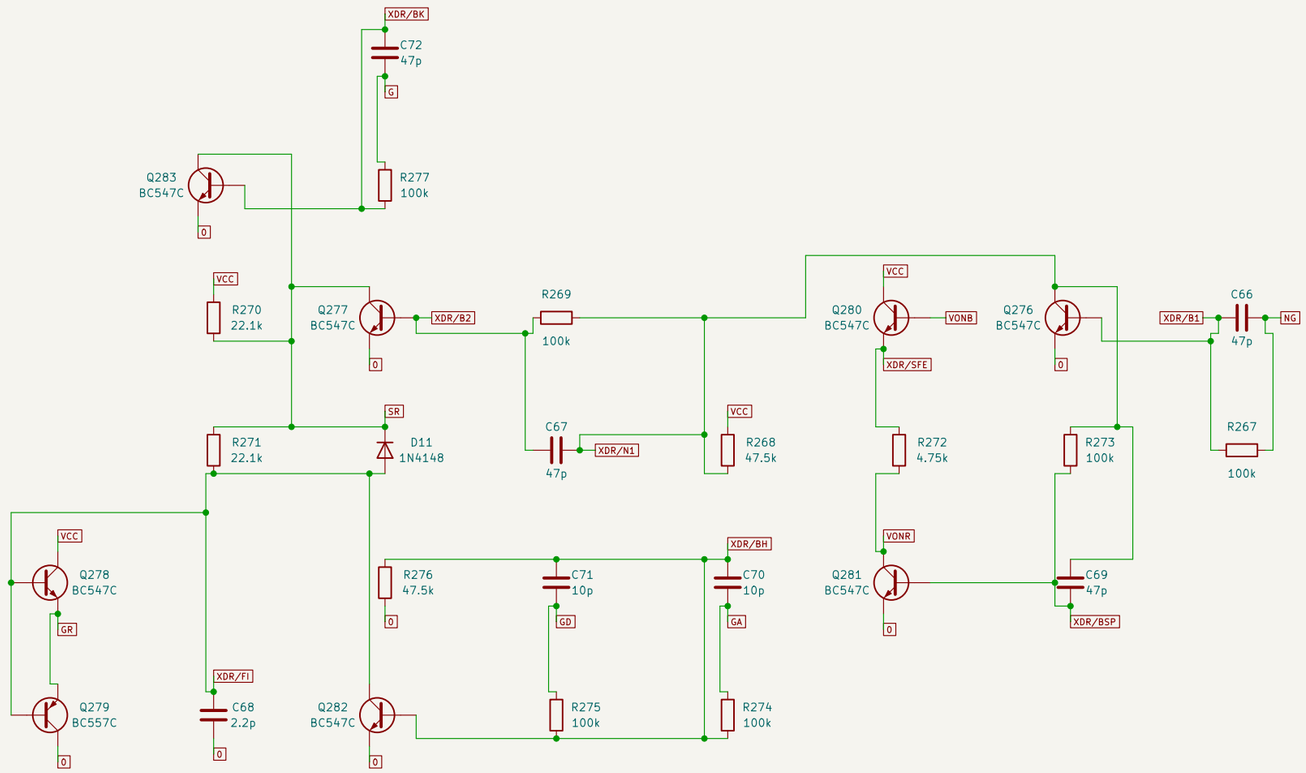
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENVO, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Gate, latch, comparator, segment logic (2)/
File: 30-gate-latch-2.kicad_sch

Title: Mamut VCA-1 A0 / Gate, latch, comparator, segment logic (2)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 17/34

Gate, latch, comparator, segment logic (3) – units XDR



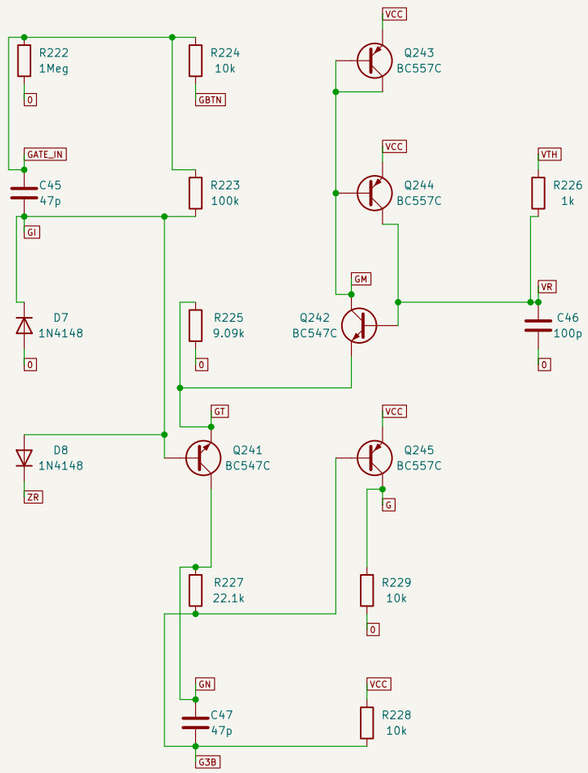
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Gate, latch, comparator, segment logic (3)/
File: 30-gate-latch-3.kicad_sch

Title: Mamut VCA-1 A0 / Gate, latch, comparator, segment logic (3)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 18/34

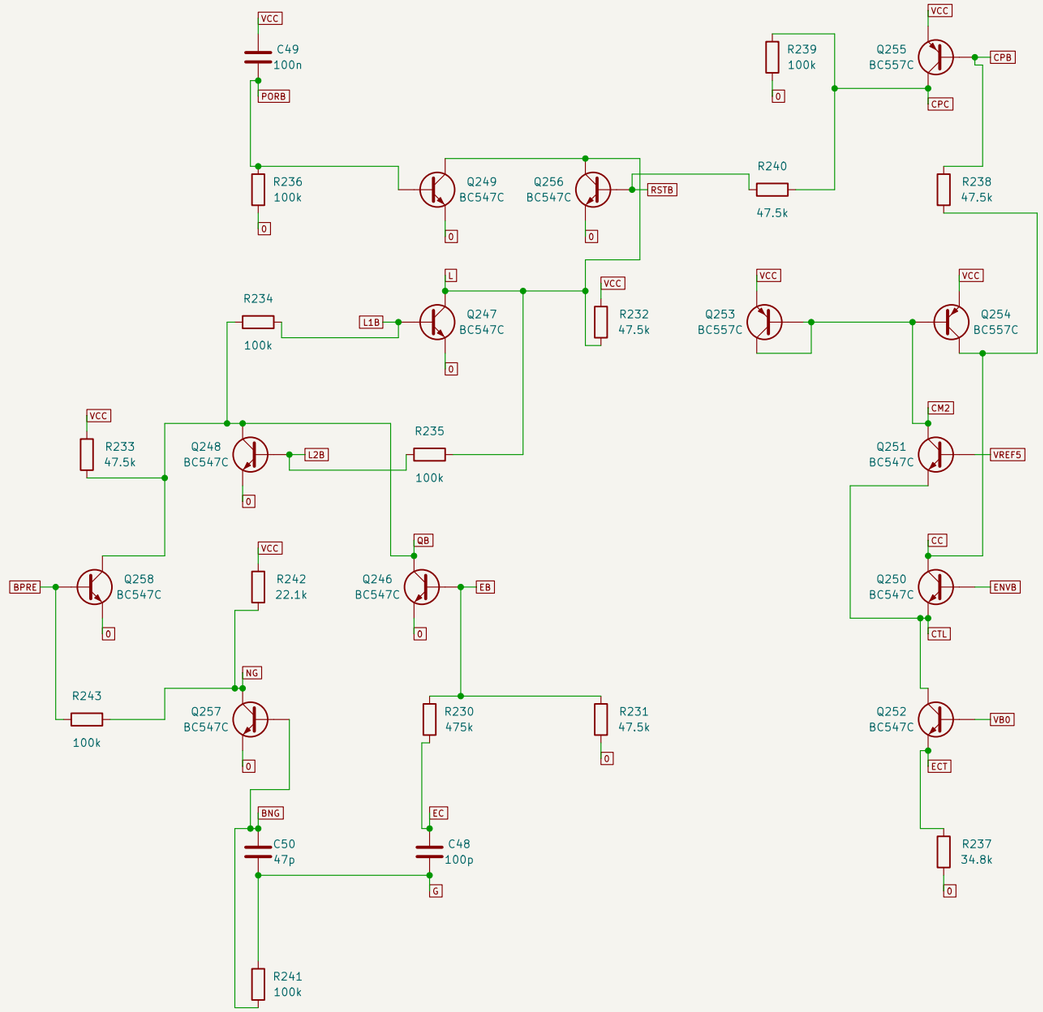
Gate, latch, comparator, segment logic (4)



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Gate, latch, comparator, segment logic (4)/		
File: 30-gate-latch-4.kicad_sch		
Title: Mamut VCA-1 A0 / Gate, latch, comparator, segment logic (4)		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 19/34

Gate, latch, comparator, segment logic (5)



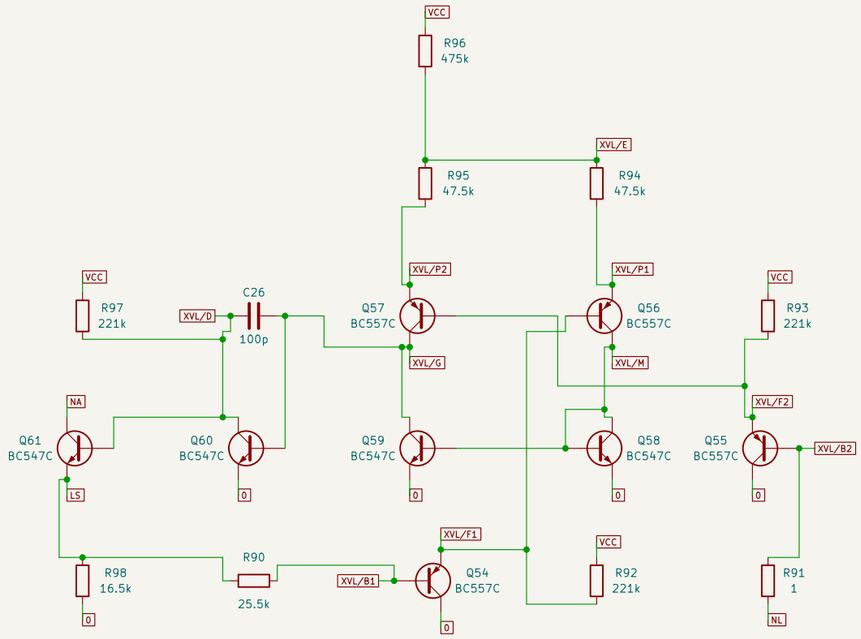
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Gate, latch, comparator, segment logic (5)/
File: 30-gate-latch-5.kicad_sch

Title: Mamut VCA-1 A0 / Gate, latch, comparator, segment logic (5)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 20/34

Control summer and current source (1) – units XVL



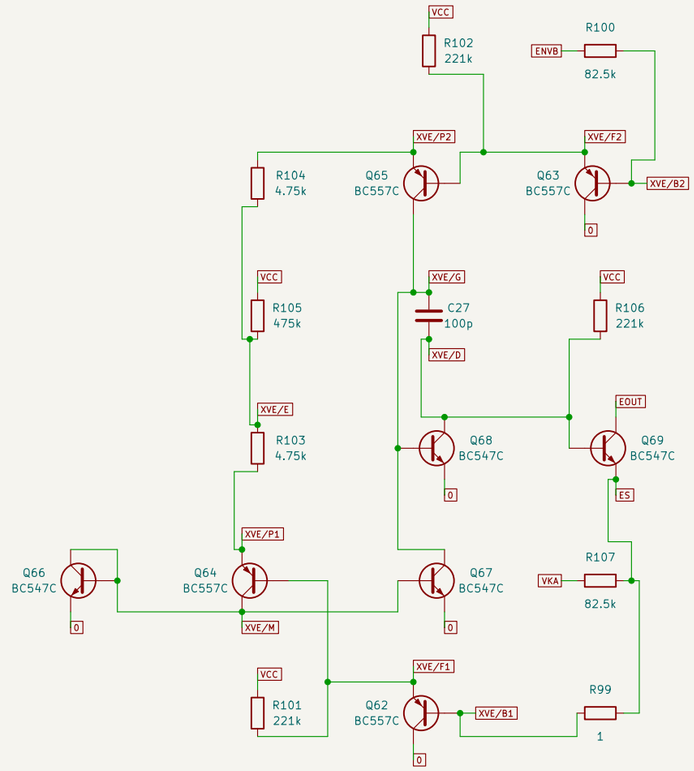
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Control summer and current source (1)/
File: 40-control-1.kicad_sch

Title: Mamut VCA-1 A0 / Control summer and current source (1)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 21/34

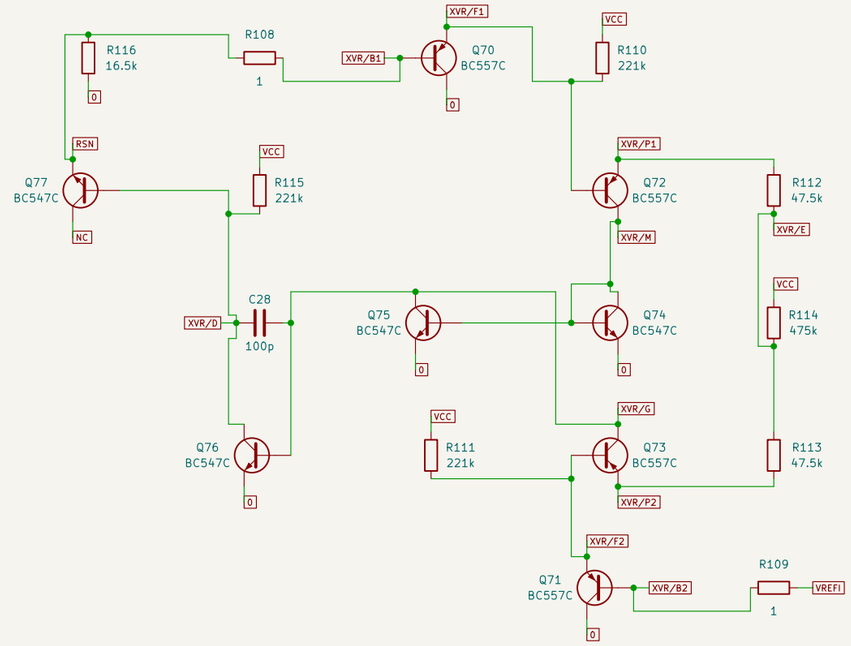
Control summer and current source (2) – units XVE



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Control summer and current source (2)/		
File: 40-control-2.kicad_sch		
Title: Mamut VCA-1 A0 / Control summer and current source (2)		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 22/34

Control summer and current source (3) – units XVR



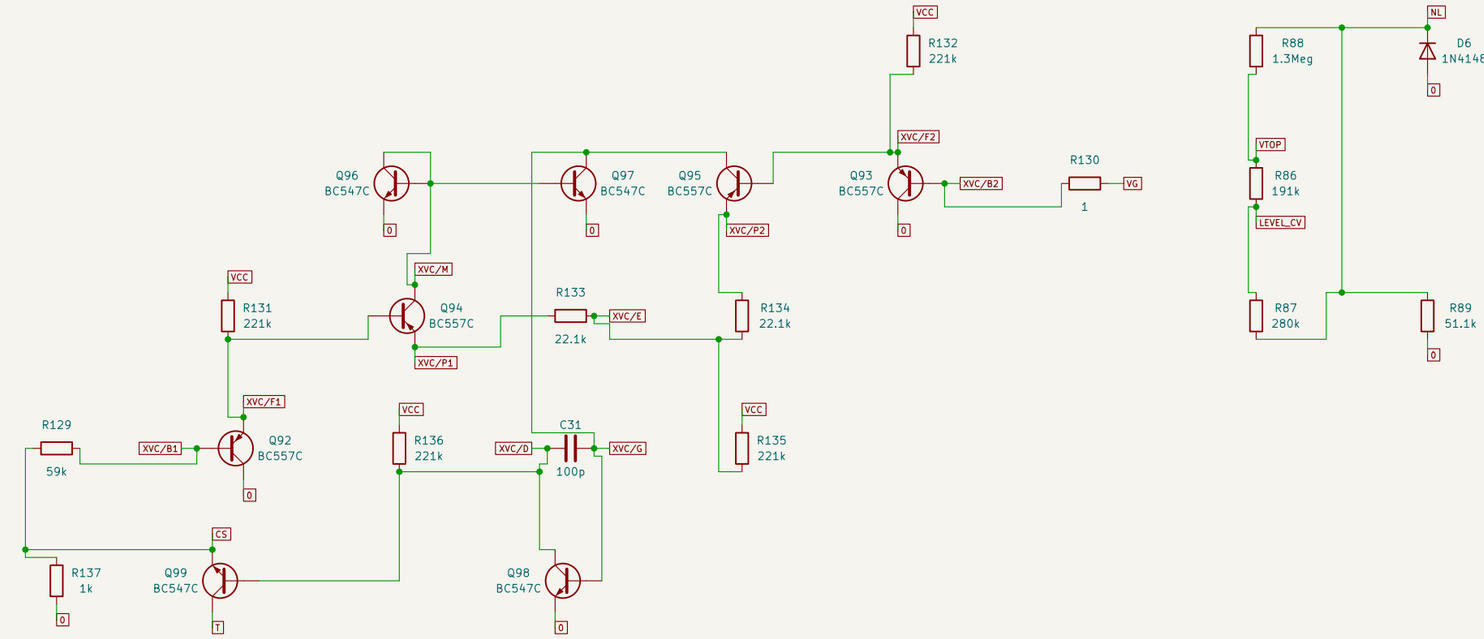
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Control summer and current source (3)/
File: 40-control-3.kicad_sch

Title: Mamut VCA-1 A0 / Control summer and current source (3)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 23/34

Control summer and current source (4) – units XVC



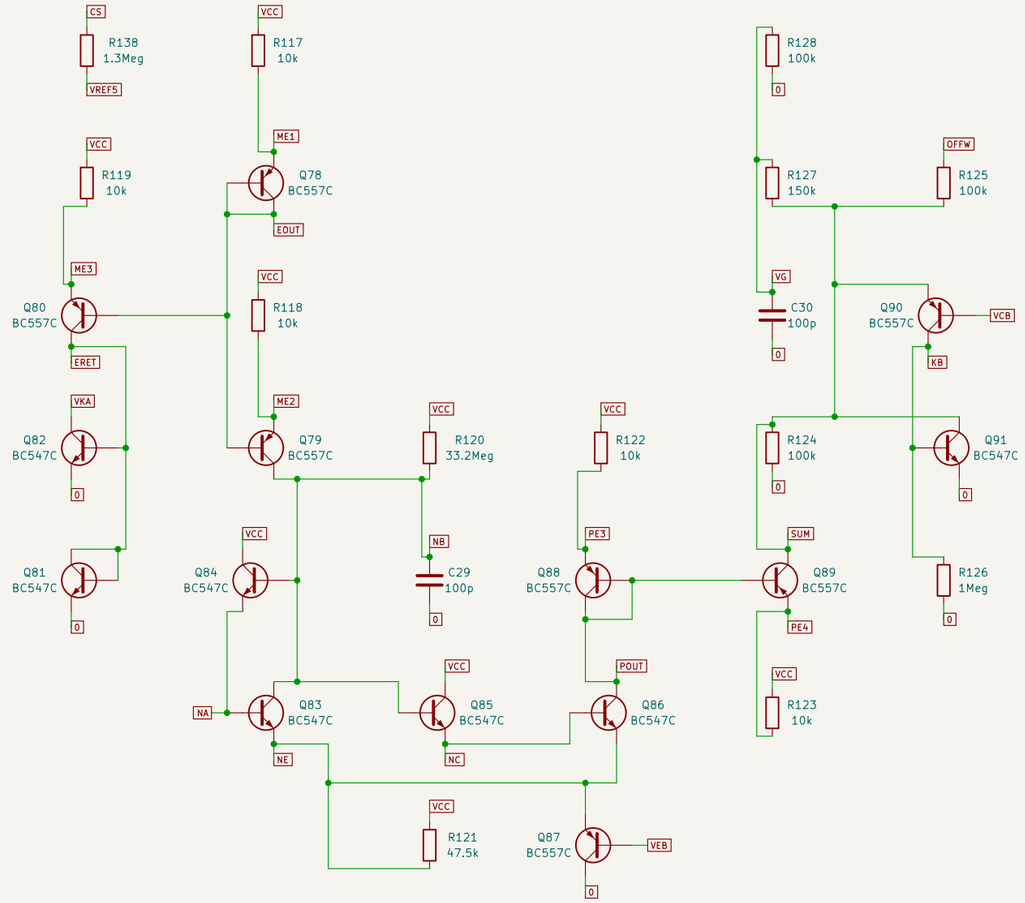
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENVO, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Control summer and current source (4)/
File: 40-control-4.kicad_sch

Title: Mamut VCA-1 A0 / Control summer and current source (4)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 24/34

Control summer and current source (5)



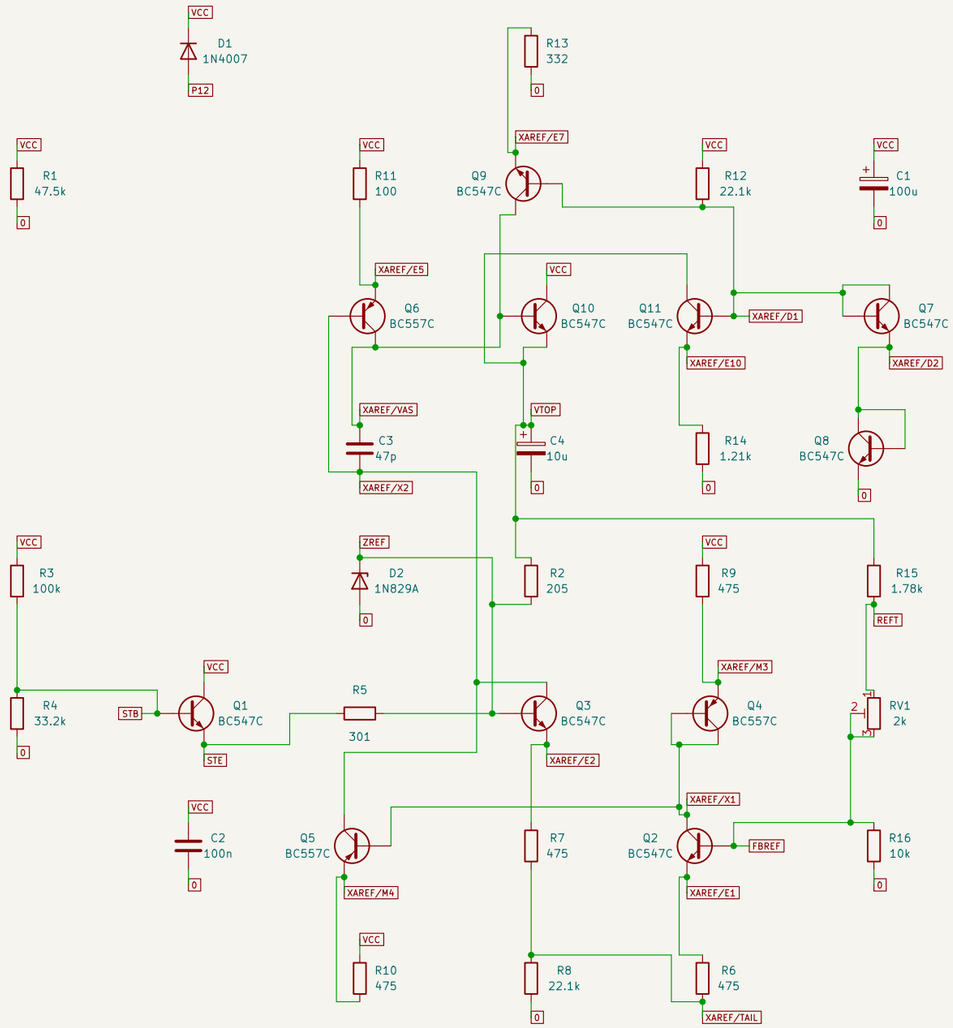
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Control summer and current source (5)/
File: 40-control-5.kicad_sch

Title: Mamut VCA-1 A0 / Control summer and current source (5)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 25/34

Power, reference, string, followers (1) — units XAREF



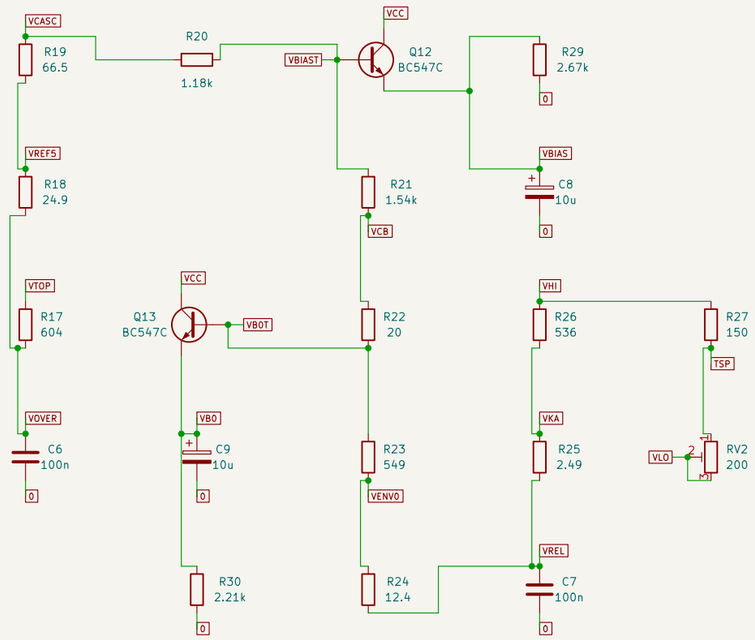
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Power, reference, string, followers (1)/
File: 50-references-1.kicad_sch

Title: Mamut VCA-1 A0 / Power, reference, string, followers (1)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 26/34

Power, reference, string, followers (2)



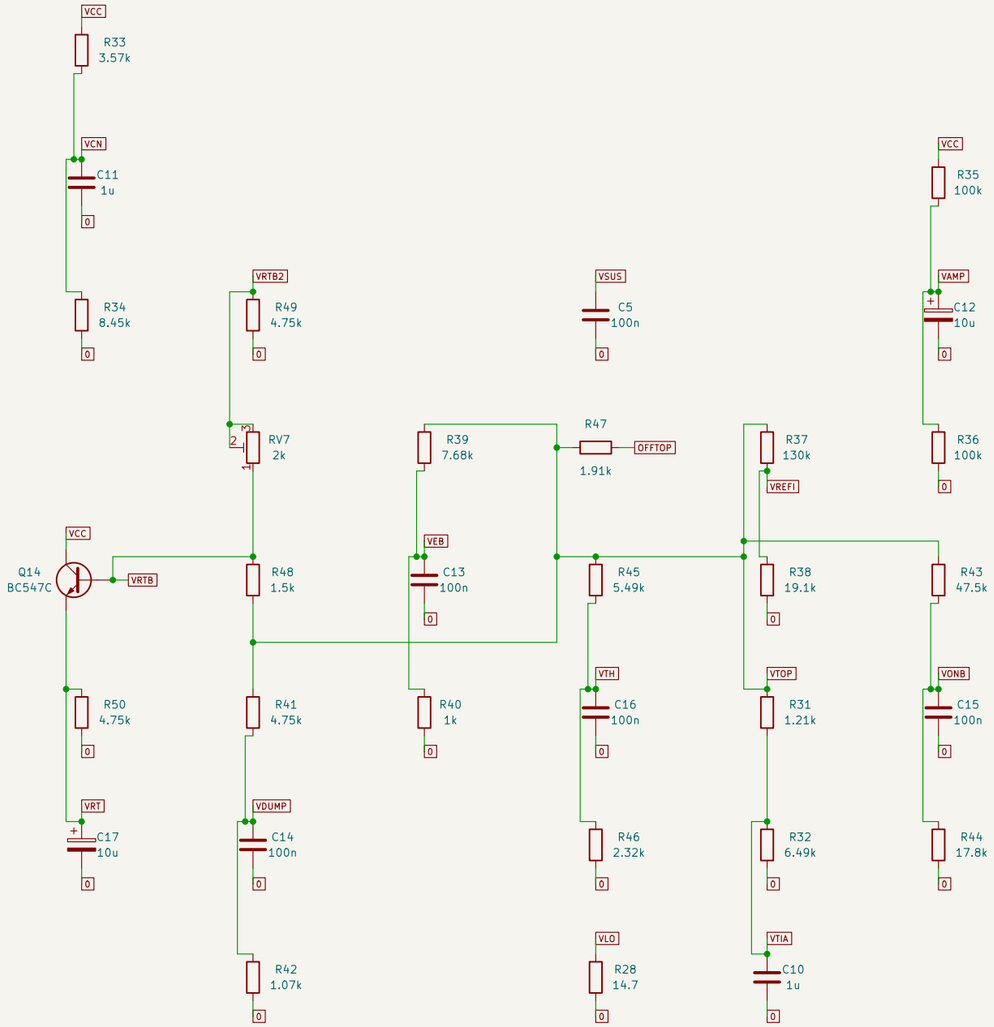
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENVO, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Power, reference, string, followers (2)/
File: 50-references-2.kicad_sch

Title: Mamut VCA-1 A0 / Power, reference, string, followers (2)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 27/34

Power, reference, string, followers (3)



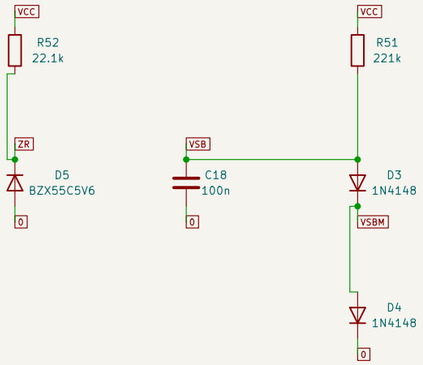
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENVO, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Power, reference, string, followers (3)/
File: 50-references-3.kicad_sch

Title: Mamut VCA-1 A0 / Power, reference, string, followers (3)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 28/34

Power, reference, string, followers (4)



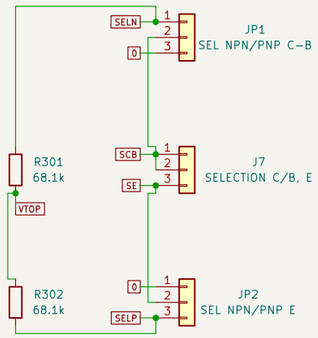
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Power, reference, string, followers (4)/
File: 50-references-4.kicad_sch

Title: Mamut VCA-1 A0 / Power, reference, string, followers (4)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 29/34

Selection position



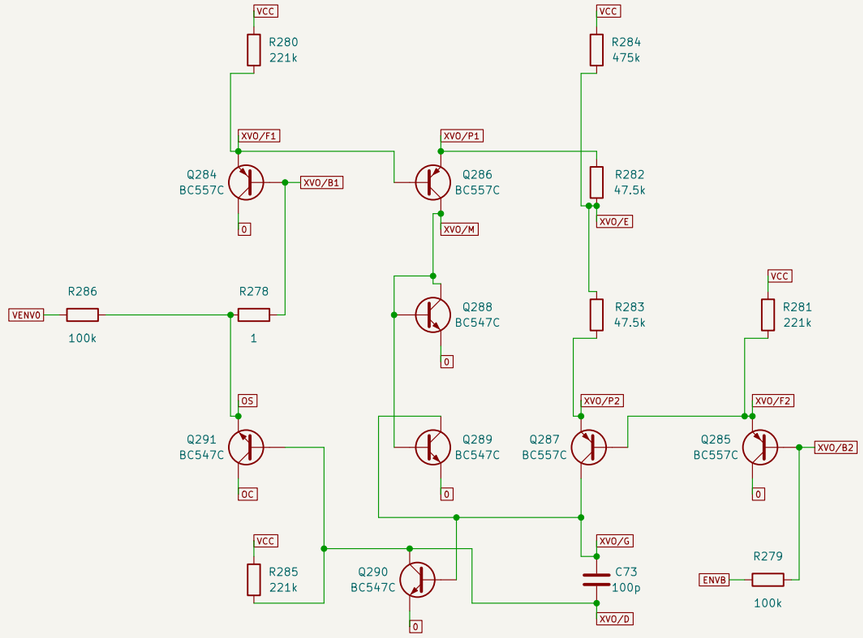
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Selection position/
File: 60-selection.kicad_sch

Title: Mamut VCA-1 A0 / Selection position

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 30/34

Controls, ENV_OUT, I/O, test points (1) — units XVO



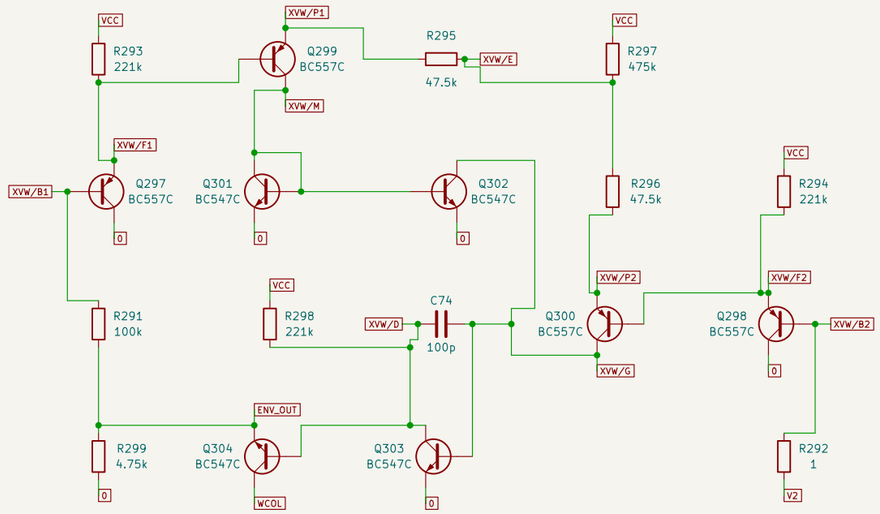
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Controls, ENV_OUT, I/O, test points (1)/
File: 70-controls-io-1.kicad_sch

Title: Mamut VCA-1 A0 / Controls, ENV_OUT, I/O, test points (1)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 31/34

Controls, ENV_OUT, I/O, test points (2) — units XVW



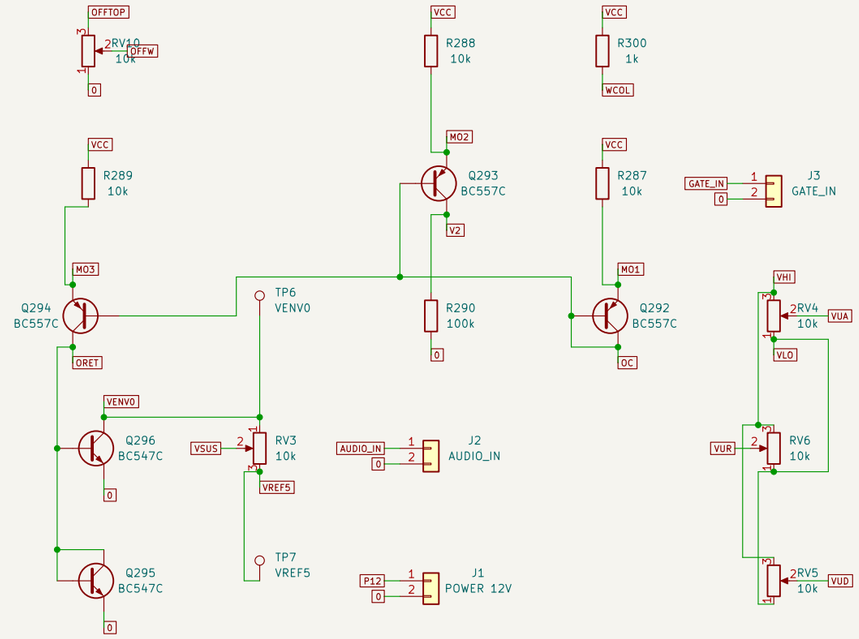
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Controls, ENV_OUT, I/O, test points (2)/
File: 70-controls-io-2.kicad_sch

Title: Mamut VCA-1 A0 / Controls, ENV_OUT, I/O, test points (2)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 32/34

Controls, ENV_OUT, I/O, test points (3)



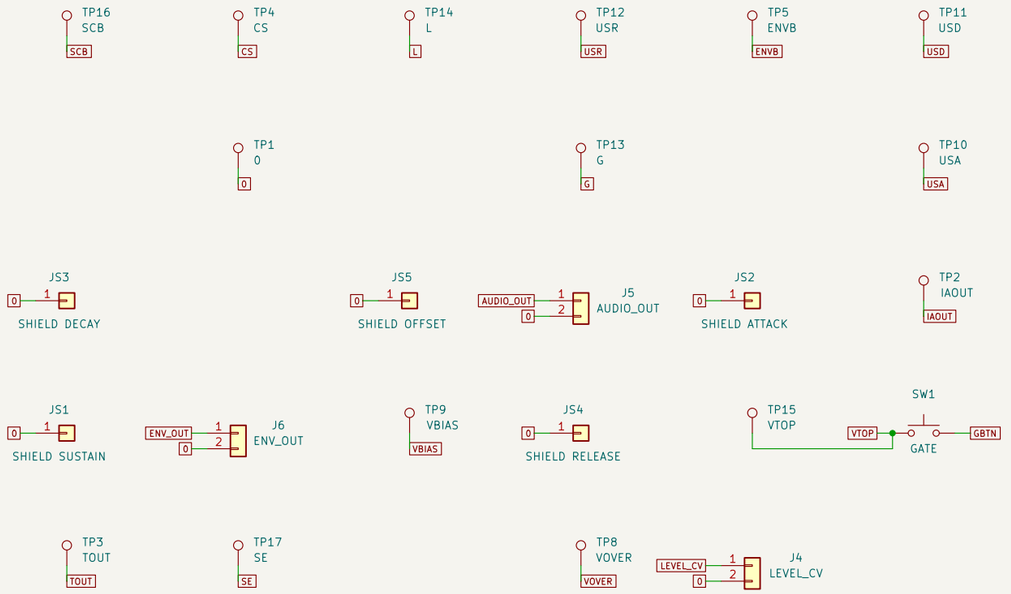
A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Controls, ENV_OUT, I/O, test points (3)/
File: 70-controls-io-3.kicad_sch

Title: Mamut VCA-1 A0 / Controls, ENV_OUT, I/O, test points (3)

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.0.7		Id: 33/34

Controls, ENV_OUT, I/O, test points (4)



A0 DEVELOPMENT / DISCRETE THT / 0 = EXTERNAL GND / VBIAS, VENV0, VREF5, VOVER = INTERNAL / PHYSICAL TESTS NOT RUN

Sheet: /Controls, ENV_OUT, I/O, test points (4)/
File: 70-controls-io-4.kicad_sch

Title: Mamut VCA-1 A0 / Controls, ENV_OUT, I/O, test points (4)

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.0.7		Id: 34/34